

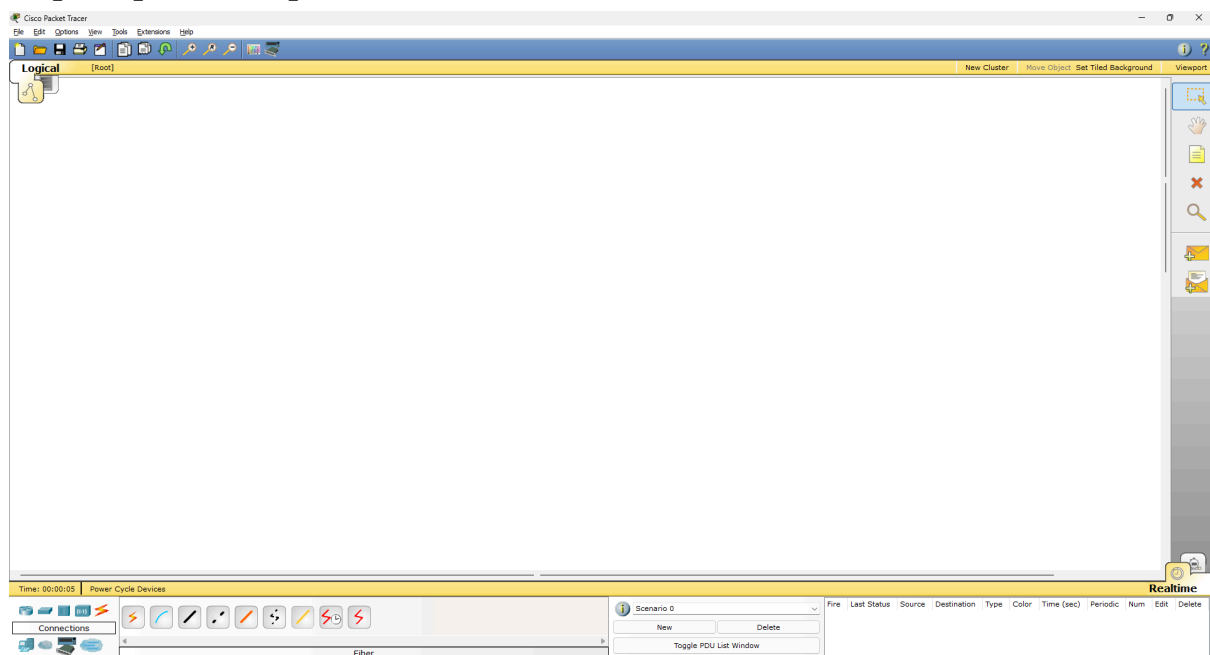
PRACTICALS

Computer Networks Laboratory

Practical 3.

Aim: To implement the topology having multiple routers (at least 3 routers) to reach the destination following static routing in packet tracer simulator.

Step 1: Open Cisco packet tracer 5.1 as shown in FIG 1

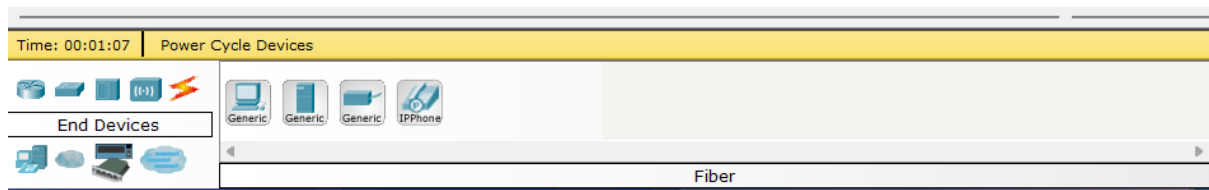


[FIG 1]

Step 2: Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in FIG 2

There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Emulation
8. Custom made device
9. Multi User connection



[FIG 2]

Step 3: Select PCs, Network Switch, & connecting wires/Cables Shown in FIG 3

1. 6 - PCs (3 for each Network)
2. 2 - Network Switch (1 For each Network)
3. 11 - Connecting wires/ cables (5 For Each Network and 1 common For both Network)
4. 2 - Server (1 For Each Network)
5. 2 - Router (1 For Each Network)

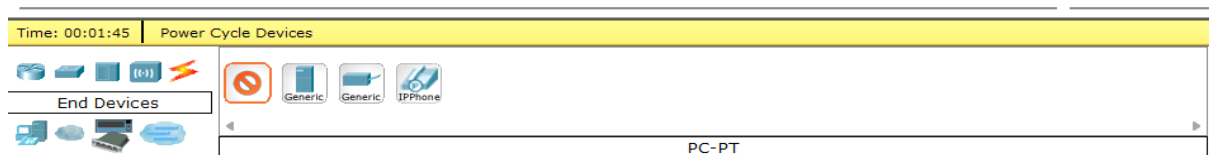


FIG 3

Step 4: Creating Network Using Required devices as shown in FIG 4

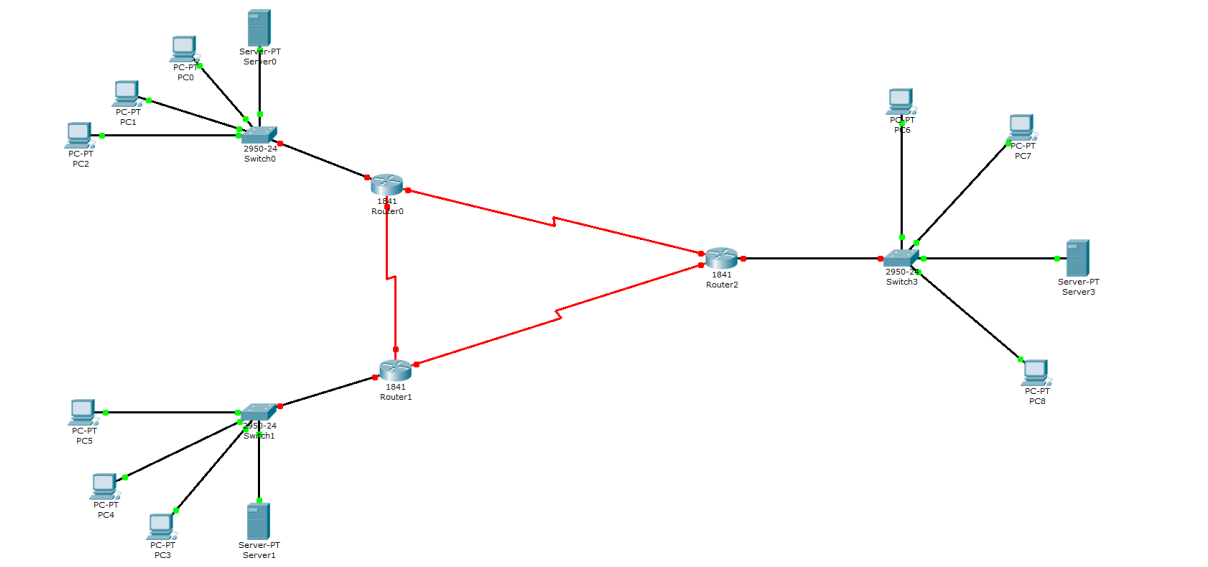


FIG 4

Step 5: Configuring The Setup with Network (DHCP method) Using Default Gateway, Subnet Mask, Ip Address as shown in FIG 5 and FIG 6

Note: This Process Is Only For One Network Do Same Process For All The Networks

1. Click on Server
2. Go to Desktop Tab
3. Click on Ip Configuration

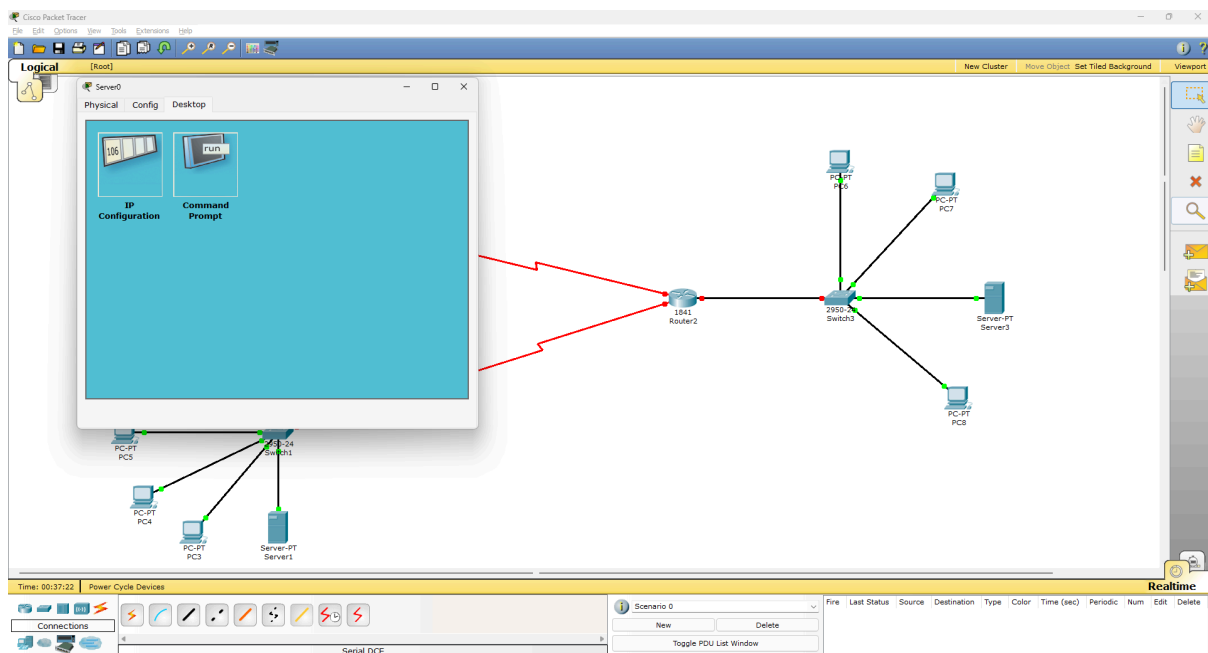
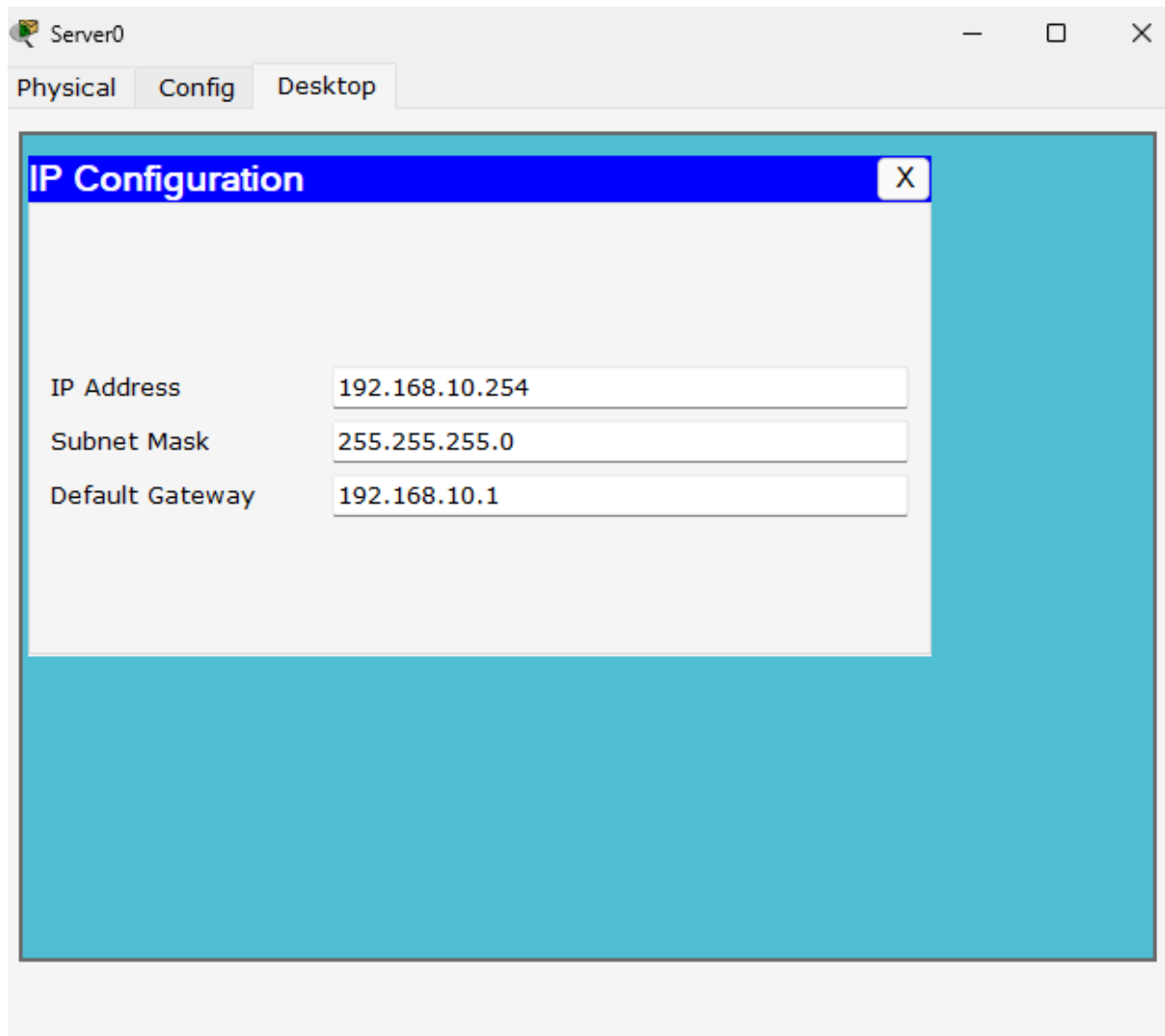


FIG 5

**FIG 6**

Step 6: Now click on PCs of Same Network Do Following Steps

1. Click on PC
2. Go to Desktop Tab
3. Click on Ip Configuration
4. Select DHCP option
5. All the configuration Details will be fetched directly from Server as Shown in FIG 7

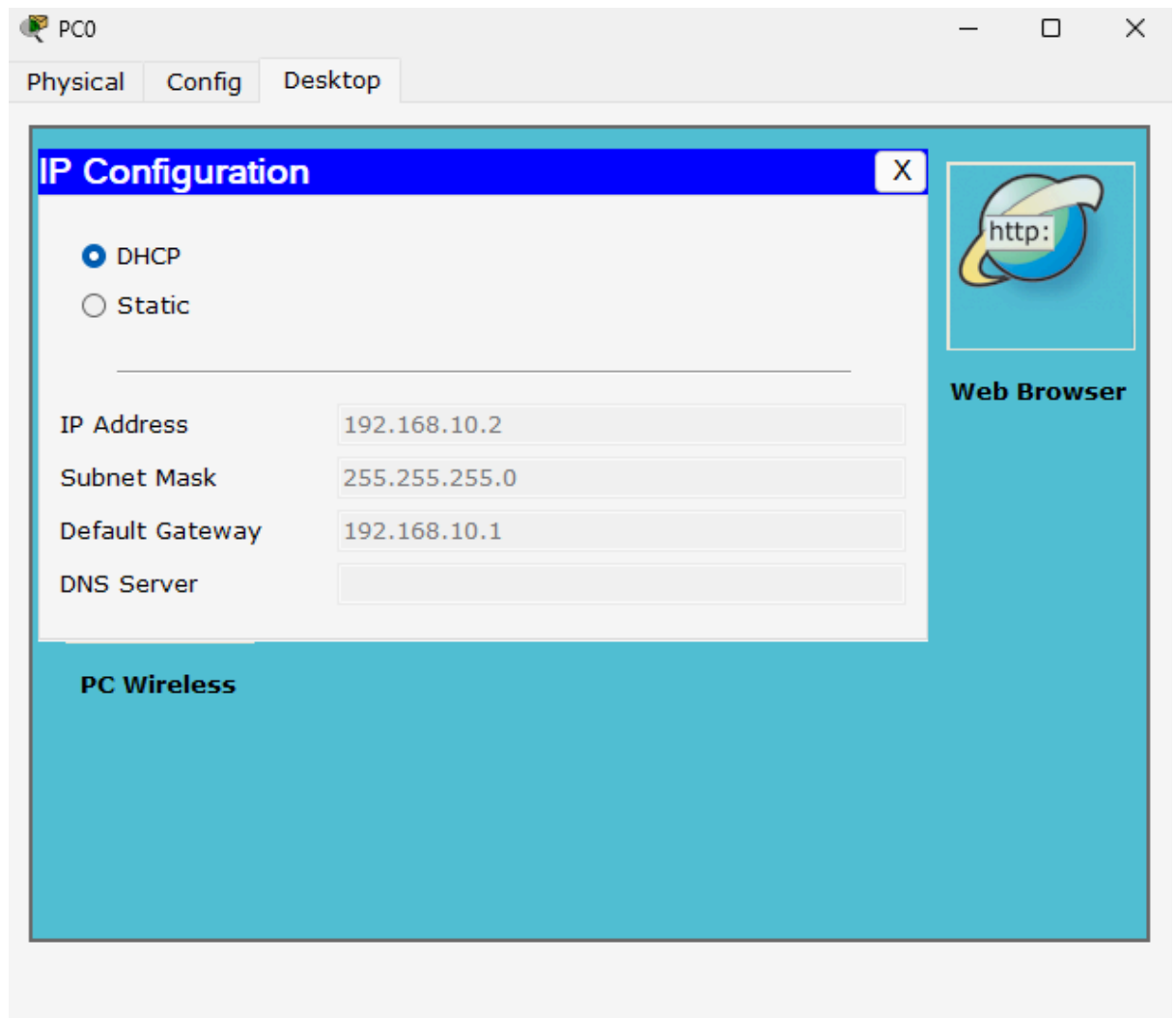
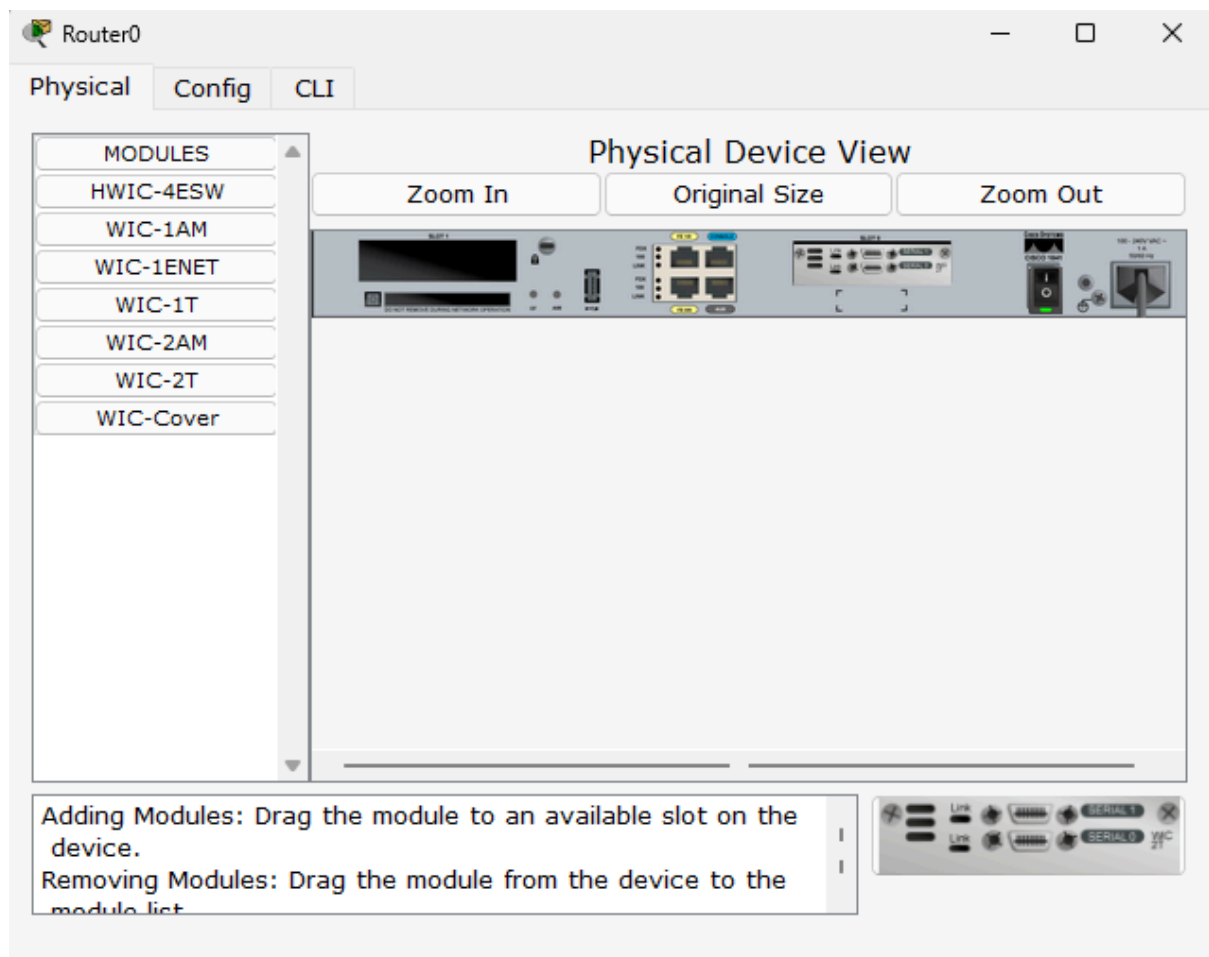


FIG 7

Step 7: Now to connect all the router we need **WIC-2T**

1. Select Router
2. Under Physical Tab you will find out **WIC-2T** Second last option
3. Turn off the switch (there is toggle button Exactly under Zoom Out button)
4. Drag and drop WIC-2T into a space between Toggle switch and Few other ports as Shown in **FIG 8**
5. Turn on the Toggle Switch
6. Do same process for all other routers

**FIG 8**

Step 8: Now Configuring Routers with the following Commands:

We have to run this commands in CLI of router, To open CLI just click on router and go to CLI and Run the Following Commands

Router1 Configuration (Network 1 side):

Open CLI of Router 1

```
enable
configure terminal
hostname Router0
```

! LAN interface (to Switch0)

```
interface FastEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

! Serial to Router1

```
interface Serial0/0/0
ip address 192.168.100.1 255.255.255.252
clock rate 64000
no shutdown
exit
```

! Serial to Router2

```
interface Serial0/0/1
ip address 192.168.100.5 255.255.255.252
clock rate 64000
no shutdown
exit
```

! Routing - OSPF advertises all connected networks

```
router ospf 1
network 192.168.10.0 0.0.0.255 area 0
network 192.168.100.0 0.0.0.3 area 0
network 192.168.100.4 0.0.0.3 area 0
exit
```

! DHCP for LAN1 (exclude gateway + server static IP)

```
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.10.254
ip dhcp pool LAN1_POOL
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 0.0.0.0
exit

end
write memory
```

Router2 Configuration (Network 2 side):**Open CLI of Router 2**

```
enable
configure terminal
hostname Router1
```

! LAN interface (to Switch1)

```
interface FastEthernet0/0
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

! Serial to Router0

```
interface Serial0/0/0
ip address 192.168.100.2 255.255.255.252
no shutdown
exit
```

! Serial to Router2

```
interface Serial0/0/1
ip address 192.168.100.9 255.255.255.252
clock rate 64000
no shutdown
exit
```

! Routing


```
router ospf 1
network 192.168.20.0 0.0.0.255 area 0
network 192.168.100.0 0.0.0.3 area 0
network 192.168.100.8 0.0.0.3 area 0
exit
```

```
! DHCP for LAN2
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.20.254
ip dhcp pool LAN2_POOL
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 0.0.0.0
exit
```

```
end
write memory
```

Router3 Configuration (Network 3 side): **Open CLI of Router 3**

```
enable
configure terminal
hostname Router2
```

! LAN interface (to Switch3)

```
interface FastEthernet0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

! Serial to Router1 (note: Se0/0/0 here pairs with R1's Se0/0/1)

```
interface Serial0/0/0
ip address 192.168.100.10 255.255.255.252
no shutdown
exit
```

! Serial to Router0

```
interface Serial0/0/1
ip address 192.168.100.6 255.255.255.252
```

```
no shutdown
exit
```

! Routing

```
router ospf 1
network 192.168.30.0 0.0.0.255 area 0
network 192.168.100.4 0.0.0.3 area 0
network 192.168.100.8 0.0.0.3 area 0
exit
```

! DHCP for LAN3

```
ip dhcp excluded-address 192.168.30.1
ip dhcp excluded-address 192.168.30.254
ip dhcp pool LAN3_POOL
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 0.0.0.0
exit
```

```
end
write memory
```

Step 9: After compilation of step 6 in the bottom Tool bar's right side we can check the status of sending packets that it's successfully Send or Failed Shown in **FIG 9.1** For Success and **FIG 9.2** For Failed

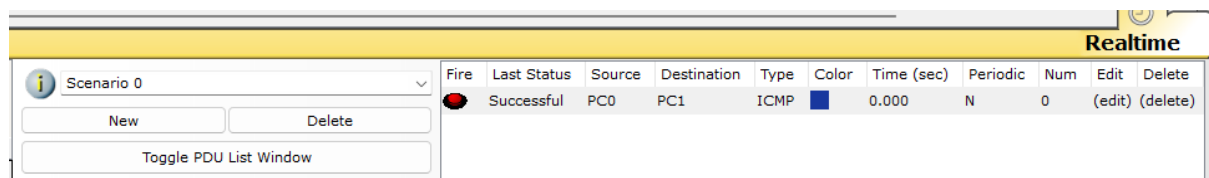


FIG 9.1

Realtime										
Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Failed	PC5	PC6	ICMP		0.000	N	0	(edit)	(delete)

FIG 9.2